

Derivación a mano:

Prop. 2:

Excedente de trabajar solo: $V^S = \omega + sH - \frac{\rho s^2}{2\pi} - b$

desarrollador procede si: $V^S \geq 0$

$$\hookrightarrow \omega + sH - \frac{\rho s^2}{2\pi} - b \geq 0 \rightarrow \omega \geq \underbrace{b - sH + \frac{\rho s^2}{2\pi}}_{T^S}$$

$$V^S \geq 0 \iff \omega \geq T^S$$

Umbral con delegación: $V^D = \omega + (1-\lambda)sH + \lambda a z(A) \dots$
 $- k(a,s) - r_D - b - \rho/2 \left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right]$

delegación es viable si: $V^D \geq 0$

$$\hookrightarrow \omega \geq (1-\lambda)sH + \lambda a z(A) - k(a,s) - r_D - b - \rho/2 [\dots]$$

$$V^D \geq 0 \iff \omega \geq T^D$$

Para un lenguaje desconocido: $\gamma^S - r_c \leq 0$

por ende: $T^C \geq T^S$

$$T^1 = \min \{T^S, T^C\} = T^S$$

Beneficio de la delegación como la reducción del

umbral: $B = T^S - T^D$

$$\hookrightarrow B = \left(b - sH + \frac{\rho s^2}{2\pi} \right) - \left[b - (1-\lambda)sH - \lambda a z(A) + k(a,s) + r_D + \rho/2 \left(\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2(a,s,A) \right) \right]$$

$$\hookrightarrow B = \lambda [a z(A) - sH] - k(a,s) - r_D + \rho/2 \left[\frac{(2\lambda - \lambda^2) s^2}{\pi} - \sigma_D^2(a,s,A) \right]$$

$$\text{si } \beta > 0 \rightarrow T^S > T^D$$

nuevo umbral: $T^2 = \min \{T^1, T^D\}$

$$T^2 = T^D < T^S = T^1$$

$$z^1 = 1 \{w > T^S\}$$

$$z^2 = 1 \{w > T^D\}$$

Consideremos 3 regiones:

$$\bullet w < T^D, \text{ no alcanza ningún umbral } z^2 = 0, z^1 = 0$$

$$z^2 - z^1 = 0$$

$$\bullet T^D \leq w < T^S \quad z^2 = 1, z^1 = 0 \quad z^2 - z^1 = 1$$

$$\bullet w > T^S \quad z^2 = 1, z^1 = 1 \quad z^2 - z^1 = 0$$

Por ende, se cambia la decisión exactamente cuando:

$$z^2 - z^1 = 1 \{T^D \leq w < T^S\}$$

Si w tiene dist. F continua: $E[z^2 - z^1] = \Pr(z^2 - z^1 = 1)$

$$\rightarrow \Pr(T^D \leq w < T^S) \rightarrow F(T^S) - F(T^D)$$

$$\rightarrow \Pr(z^2 - z^1 = 1) = F(T^S) - F(T^D)$$